

GROUP 1

CASE STUDY 2026

ANALYSIS BASED ON THE 2010 UK ETHNIC MINORITY POST-ELECTION SURVEY

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RESEARCH QUESTION

Does Annual Household Income Play a Role in Minority's Political Party Identification?

This research question is of interest to several key stakeholders. Political parties would benefit from understanding how income shapes party allegiance among ethnic minority voters, allowing them to tailor their policies and outreach more effectively. Policymakers would find this valuable because it reveals how socioeconomic conditions which are directly shaped by government policies, influence political behaviour within minority communities. Finally, local authorities and constituency representatives would gain insight into what issues matter most to the communities they serve, helping them better evaluate how economic policies impact political behaviour.

DATASETS USED

The analysis uses data from the British Election Study (BES) Ethnic Minority Survey 2010. The relevant files were located in the stata folder of the dataset package.

Data File:

- BSE_2010.dta
- This is the Stata dataset containing responses from 2,787 participants and 637 variables.

Documentation Files:

- Data Dictionary (RTF format)
- The data dictionary provides detailed information about each variable, including:
 - Variable names and labels
 - Value labels and coding schemes
 - Measurement levels
 - Missing-value definitions

The data dictionary is used to identify the variables required for the analysis and to convert coded survey responses into meaningful categorical labels. For this case, we made use of Python's 'pyreadstat' library that auto reads the encoded dictionary embedded in the .dta file and can retrieve that as a 'meta variable' alongside the data frame (df) variable (see Python notebook attached)

For this study, the key variables were:

- zqinc – Annual Household Income
- bq9_1 – Political Party Identification

These variables were extracted from the dataset and their value labels were obtained from the dictionary with the '*meta.variable_value_labels*' method.

VARIABLES OF INTEREST

Our two variables of interest are

Income Groups (zqinc): it is a categorical variable of ordinal type because the income bands are ordered based on the numerical value. However, the variable is still categorical as they are ranges of numerical numbers quoted as strings (non-numerical).

Party identification (bq9_1): political party label, which is again a categorical type of variable. However, it is a nominal categorical variable with no apparent order or ranking between the parties.

The categories are encoded as numbers and their meaning are indicated below for both the variables of interest in this analysis

I. Annual Household Income (Ordinal)—zqinc:

- 1 = £0 – 5,000
- 2 = £5,001 – 10,000
- 3 = £10,001 – 15,000
- 4 = £15,001 – 20,000
- 5 = £20,001 – 25,000
- 6 = £25,001 – 30,000
- 7 = £30,001 – 35,000
- 8 = £35,001 – 40,000
- 9 = £41,000 – 45,000
- 10 = £45,001 – 50,000
- 11 = £50,001 – 60,000
- 12 = £60,001 – 70,000
- 13 = £70,001 – 80,000
- 14 = £80,001 or more

II. Party Identification (Nominal)—bq9_1

- 1 = None/No party
- 2 = Labour
- 3 = Conservatives
- 4 = Liberal Democrats
- 5 = Scottish National Party
- 7 = Green Party
- 8 = UKIP
- 9 = BNP
- 11 = Other
- 12 = Respect

A few notable points in this context:

- Values -2 and -1 ('Refused' and 'Don't Know') were excluded from analysis. More on this in the 'Assumptions' section below.
- People who participated in the survey but specifically chose not to answer the income question cannot be considered in the analysis
- Similarly, people who refused to answer Party question, either didn't want to reveal their political leaning or genuinely hadn't formed a party preference.

ANALYSIS METHODS

We used the Chi-square test of independence which is essentially designed for categorical variables. To establish whether the distribution of one variable differs significantly across the categories of another, a chi-square test is performed which also gives out a p value determining how meaningful the pattern might be. In this case, the two categorical variables are annual household income and political party identification

Chi square test is mainly designed for a contingency table comparing two groups of variables that are mainly categorical or non-numerical. Since we need to understand if there is an existing pattern in the distribution of these two categories, we can estimate that by constructing a contingency table crossing both variables and comparing the observed frequencies against what we would expect if the two variables were completely unrelated.

ASSUMPTIONS

The chi-square test of independence requires the following assumptions to be met:

1. Both variables must be categorical. Annual household income is ordinal categorical (as mentioned above) and party identification is nominal categorical. This assumption is satisfied.
2. The Chi-square test requires that the observations are independent. Each respondent appears only once in the dataset. This assumption is satisfied as each row represents a unique survey participant associated with a Unique ID listed in the first column of the dataset.
3. For Chi-square tests to be relevant, the expected frequencies in each cell of the contingency table should be at least 5, with no more than 20% of cells falling below this threshold. Meaning at least 80% of the cells must have a value >5 . When we initially ran the test using all 14 income bands, 34.3% of cells violated this assumption (see Python notebook attached). To address this, we grouped the 14 income bands into 4 broader categories as follows: Low (under £20k), Lower-middle (£20k-£35k), Upper-middle (£35k-£60k) and High (over £60k). After regrouping, only 10% of cells fell below the threshold, satisfying the assumption.
4. Respondents who refused to answer or said they don't know for either variable were excluded (1,190 respondents, 43% of the original sample). This is a limitation as these respondents may have different political patterns that cannot be measured.

NULL AND ALTERNATIVE HYPOTHESIS STATEMENTS

Null Hypothesis (H_0):

Annual household income and political party identification are independent of each other among ethnic minority respondents. In other words, there is no association between household income and party identification.

Alternative Hypothesis (H_1):

Annual household income and political party identification are associated among ethnic minority respondents. In other words, political party identification varies according to household income levels.

Statistical Test:

A Chi-square test of Independence is used here to determine whether there is a statistically significant association between income category and political party identification.

Decision Rule:

- If $p\text{-value} < 0.05$: Reject H_0 and conclude that a significant association exists.
- If $p\text{-value} \geq 0.05$: Fail to reject H_0 and conclude that there is insufficient evidence of an association.

RESULTS

Chi-squared Results:

Chi-square statistic: 35.8038

Degrees of freedom: 12

P-value: 0.0003

Looking at the results, we see that the p-value (0.0003) falls below the level of significance (0.05). This leads us to reject the null hypothesis and therefore conclude that there is a statistically significant association between annual household income and minority's party identification.

Looking at Figure 1 below we observe a notable increase in the identification with the Conservative party as annual household income increases. In addition, high income has the highest percentage of individuals who answered None/No.

Overall, identification with the Labour Party is dominant in each household income category however, we see a gradual increase from 69.5% among low-income respondents to 55.7% among high income respondents in the identification of this party as household income increases.

Support for the Liberal Democrats Party increases as annual household income increases from Low to Upper Middle then takes a sudden decrease in the High-income category.

Lastly when we look at the data representing Other, we see that the Upper Middle-income category has the lowest percentage supporting other parties.

Proportional Party Identification Within Annual Household Income Categories (UK 2010)

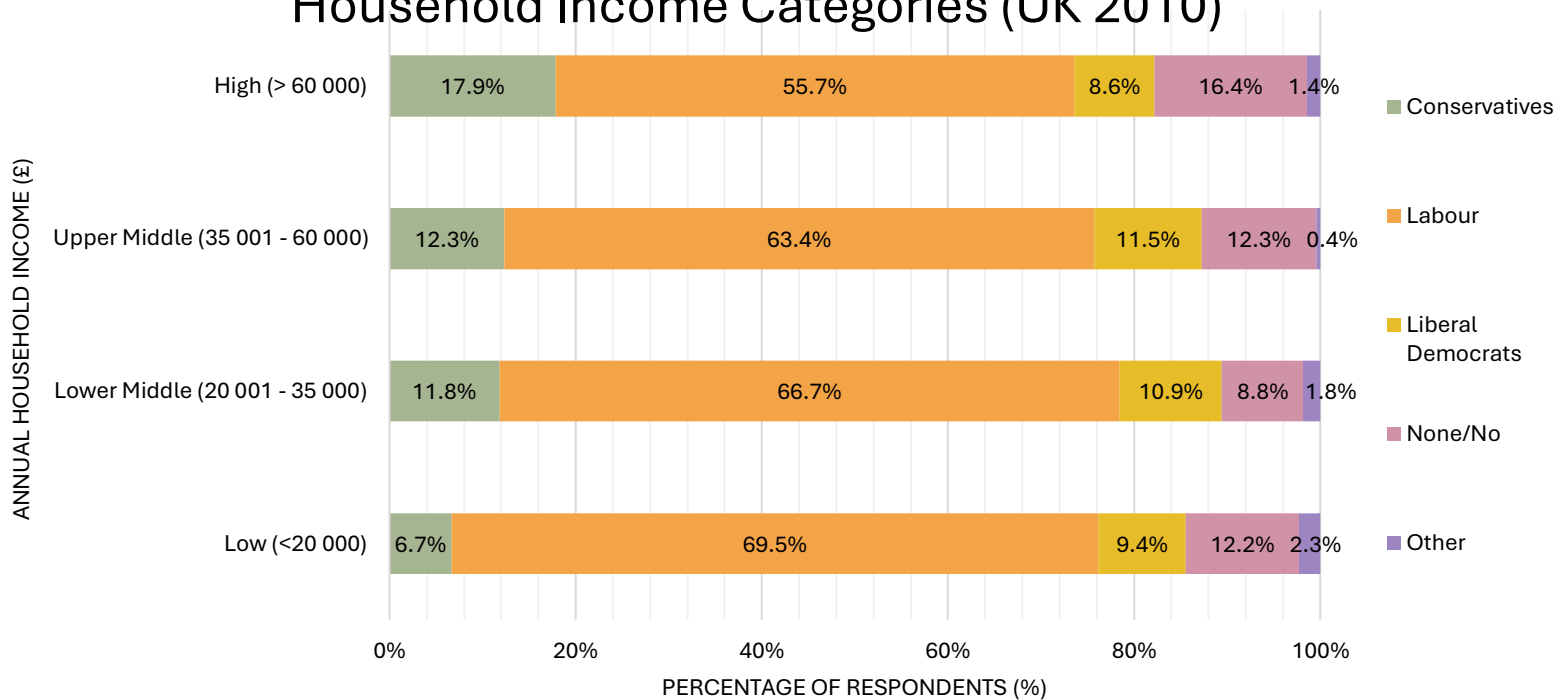


Figure 1: 100% stacked bar graph displaying the variation of party identification within different annual household income categories

The use of a percentage stacked and not a standard stacked bar graph was since the different annual household income categories each have different total counts of party identification. For example, the total count of Low income is 884 and the total count of High income is 140. Using the 100% stacked bar graph allows us to strictly view the variation of party identification within each individual income category.

IMPACT

The results of this study provide strong evidence against the existence of monolithic, or bloc, voting patterns. The use of bloc voting is a long-standing political assumption, thus providing mathematical evidence that challenges this assumption is highly impactful.

A meaningful insight gathered from this study is the difference in support toward the Conservative party across financial brackets. The higher the annual income of a household, the higher probability there is of them identifying with the Conservative Party. A study done in 2010 by Jo Thori Lind shows that there is a positive relationship between individuals who vote conservative and those who wish tax to reduce. This information supports the idea that minorities are voting conservative as they gain wealth to support preserving it. This shows that minorities are supporting parties based on economics and less on identity.

The data shows that different income households have different priorities and lean in different directions. An insight such as this has a significant impact on the decisions of policymakers tasked with designing targeted, effective legislation. These policies could be targeted towards financial safety nets such as financial aid at universities and increasing minimum wage.

REFERENCES

- Heath, A. et al., British Election Study Ethnic Minority Survey, 2010 [computer file]. Colchester, Essex: UK Data Archive, April 2012. SN: 6970. <http://dx.doi.org/10.5255/UKDA-SN-6970-1>
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